

15-mavzu. Egri chiziq egriligi va uni hisoblash. Chiziq burailishi va uni hisoblash. Frene formulalari. Chiziqning tabiiy tenglamalari.

$$\begin{aligned} S &= \int_a^{3a} \sqrt{1^2 + \frac{t^4}{a^4} + \frac{a^4}{4t^4}} dt = \int_a^{3a} \sqrt{\frac{4a^4 t^4 + 4t^8 + a^8}{4a^4 t^4}} dt = \int_a^{3a} \frac{\sqrt{(a^4 + 2t^4)^2}}{2a^2 t^2} dt = \\ &= \int_a^{3a} \frac{a^4 + 2t^4}{2a^2 t^2} dt = \frac{a^2}{2} \int_a^{3a} \frac{dt}{t^2} + \frac{1}{a^2} \int_a^{3a} t^2 dt = \frac{a^2}{2} \left(-\frac{1}{t} \right) \Big|_a^{3a} + \frac{1}{a^2} \left(\frac{t^3}{3} \right) \Big|_a^{3a} = \\ &= -\frac{a}{6} + \frac{a}{2} + 9a - \frac{a}{3} = \frac{-a + 3a - 2a}{6} + 9a = 9a. \end{aligned}$$

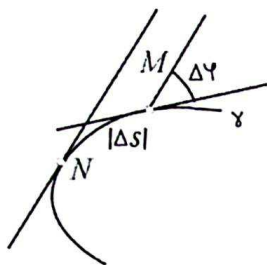
§ 6. Egri chiziq egriligi va uni hisoblash

Bizga regulyar γ –egri chiziq va M unga tegishli nuqta berilgan bo'lsin. Berilgan M nuqtadagi egrilik tushunchasini kiritib, uni hisoblash formulasini keltirib chiqaramiz. Buning uchun γ – egri chiziqda M ga yaqin bo'lgan N nuqtani olib, bu nuqtalardan o'tuvchi urinmalar orasidagi burchakni $\Delta\varphi$ bilan, $\overset{\circ}{MN}$ yoy uzunligini ΔS bilan belgilaylik. Ravshanki, N nuqta M ga intilganda $\Delta\varphi$ va ΔS miqdorlar nolga intiladi. Ammo $\frac{\Delta\varphi}{\Delta S}$ ifoda nimaga intilishini oldindan ayta olmaymiz.

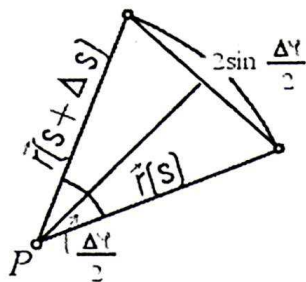
Ta'rif. Chiziqdagi N nuqta M ga intilganda $\frac{\Delta\varphi}{\Delta S}$ ifodaning limiti mavjud bo'lsa, u γ chiziqning M nuqtadagi **egriligi** deb ataladi.

Teorema-12: Ikki marta differentsiallanuvchi regulyar egri chiziq uchun $k = \lim_{M \rightarrow N} \frac{\Delta\varphi}{\Delta S}$ mavjud. Agar γ chiziq $\vec{r} = \vec{r}(s)$ tenglama bilan

tabiiy parametr yordamida berilgan bo'lsa, $k = \left| \vec{r}'(s_0) \right|$ tenglik o'rinlidir. Bu yerda s_0 tabiiy parametrning M ga mos keluvchi qiymatidir.



Chizma-9



Chizma-10

Isbot. Faraz qilaylik, γ egri chiziq $\vec{r} = \vec{r}(s)$ tenglama bilan tabiiy parametr yordamida berilgan, $\vec{r}(s_0)$, $\vec{r}(s_0 + \Delta s)$ vektorlar mos ravishda M va N nuqtalarning radius vektorlari bo'lsin. Shunda $\Delta\varphi$ burchak $\vec{r}(s_0)$ va $\vec{r}(s_0 + \Delta s)$ vektorlar orasidagi burchakka teng.

Shuning uchun $|\vec{r}(s_0 + \Delta s) - \vec{r}(s_0)| = 2 \sin \frac{\Delta\varphi}{2}$. Bu tenglikdan,

$$\frac{|\vec{r}(s_0 + \Delta s) - \vec{r}(s_0)|}{\Delta s} = \frac{2 \sin \frac{\Delta\varphi}{2}}{\Delta s} = \frac{\sin \frac{\Delta\varphi}{2}}{\frac{\Delta\varphi}{2}} \cdot \frac{\Delta\varphi}{\Delta s}$$

kelib chiqadi. Bu tenglikda $\Delta s \rightarrow 0$ da limitga o'tsak, $k = |\ddot{\vec{r}}(s)|$ ni hosil qilamiz. $\square$

Endi ixtiyoriy parametr uchun egrilikni hisoblash formulasini keltirib chiqaramiz. Buning uchun $\vec{r} = \vec{r}(s)$ tenglikda s ni t ning funksiyasi sifatida qarab, ikkala tomonini t bo'yicha differensiallaylik. Shunda $\dot{\vec{r}} = \dot{\vec{r}}s'$ ni hosil qilamiz. Demak,

$$\dot{\vec{r}} = \frac{\vec{r}'}{|\vec{r}'|}.$$

Endi bu tenglikni t bo'yicha differensiallaymiz va

$$\dot{\vec{r}}s' = \frac{\vec{r}''|\vec{r}'| - (\vec{r}', \vec{r}'')}{\sqrt{(\vec{r}', \vec{r}')}} \cdot \frac{1}{|\vec{r}'|^2}$$

ni hosil qilamiz. Bu tenglikni ikkala tomonini s' ga bo'lib

$$\ddot{\vec{r}} = \frac{\vec{r}''}{|\vec{r}'|^2} - \frac{(\vec{r}', \vec{r}'')\vec{r}'}{|\vec{r}'|^3}$$

tenglikni olamiz. Endi ikkala tomonini kvadratga oshirib,

$$k^2 = \frac{\vec{r}'''^2 \vec{r}'^2 - (\vec{r}', \vec{r}'')^2}{|\vec{r}'|^6}$$

tenglikni hosil qilamiz.

Bundan esa $k = \frac{|[\vec{r}', \vec{r}'']|}{|\vec{r}''|^3}$ kelib chiqadi.

Bu ifodada $|\vec{r}'| = \sqrt{(\vec{r}', \vec{r}')} = \sqrt{\vec{r}'^2}$ tenglikni hisobga olib formulani ko'rinishda

$$k = \frac{|[\vec{r}', \vec{r}'']|}{|\vec{r}'^2|^{\frac{3}{2}}}$$

yozib ixtiyoriy parametr uchun egrilikni hisoblash formulasini olamiz.

Agar $\vec{r}(t) = \{x(t), y(t), z(t)\}$ bo'lsa, formula

$$k = \frac{\sqrt{\begin{vmatrix} y' & z' \\ y'' & z'' \end{vmatrix}^2 + \begin{vmatrix} z' & x' \\ z'' & x'' \end{vmatrix}^2 + \begin{vmatrix} x' & y' \\ x'' & y'' \end{vmatrix}^2}}{(x'^2 + y'^2 + z'^2)^{\frac{3}{2}}}$$

ko'rinishiga keladi. Agar γ egri chiziq $y = f(x)$ funksiyani grafigi bo'lsa, egrilik formulasi

$$k = \frac{|f''|}{(1 + f'^2)^{\frac{3}{2}}}$$

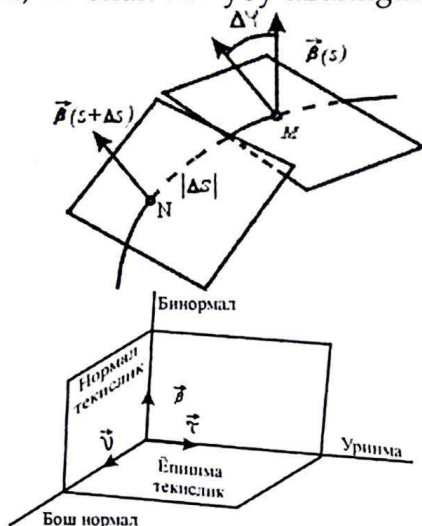
ko'rinishga keladi.

Endi, hamma nuqtalarida egriligi nolga teng bo'ladigan chiziqlarni topaylik. Ikki marta differensiallanuvchi egri chiziq tabiiy parametr yordamida $\vec{r} = r(s)$ tenglama yordamida berilgan bo'lsa, uning egriligi formula $k = |\ddot{\vec{r}}(s)|$ bo'yicha hisoblanadi. Agar $k = 0$ bo'lsa, $|\ddot{\vec{r}}(s)| = 0$ bo'ladi. Demak, $|\ddot{\vec{r}}(s)| = 0$ va

$\vec{r}(s) = \vec{a}s + \vec{b}$ bo'lib, $\vec{a}, \vec{b}$ – o'zgarmas vektorlardir. Demak, egri chiziqning hamma nuqtalarida egriligi nolga teng bo'lsa, u yoki to'g'ri chiziq, yoki to'g'ri chiziqning ochiq kesmasidir. Albatta, bu tasdiqning teskarisi ham to'g'ridir (isbotlang).

§ 7. Egri chiziqning buralishi va uni hisoblash

Egri chiziqning berilgan M nuqtasidagi buralishi tushunchasini kiritaylik. Bizga γ egri chiziq va unga tegishli M nuqta berilgan bo'lsin. M nuqtaga yaqin va γ ga tegishli nuqtani N bilan, $\Delta\varphi$ bilan bu nuqtalardan o'tuvchi yopishma tekisliklar orasidagi burchakni, Δs bilan $\overset{\frown}{MN}$ yoy uzunligini belgilaylik.



Chizma-11

Chizma-12

Ta'rif: N nuqta M nuqtaga intilganda $\frac{\Delta\varphi}{\Delta s}$ ifodaning limiti γ egri chiziqning M nuqtadagi *absolyut buralishi* deyiladi va $|\sigma|$ bilan belgilanadi.

Teorema-13: Uch marta differensiallanuvchi regulyar γ egri chiziqligining, M nuqtada egriligi noldan farqli bo'lsa, $\frac{\Delta\varphi}{\Delta s}$ ifoda tayin limitga ega. Agar γ egri chiziq tabiiy parametr yordamida

$$\vec{r} = \vec{r}(s)$$

tenglama bilan berilgan bo'lsa, uning absolyut buralishi,

$$|\sigma| = \frac{|\dot{\vec{r}} \ddot{\vec{r}} \ddot{\vec{r}}|}{|\ddot{\vec{r}}|^2}$$

formula bo'yicha hisoblanadi.

Isbot: Faraz qilaylik, M nuqtadagi egrilik noldan farqli bo'lsin. Egrilik uzluksiz funksiya bo'lganligi uchun M ga yaqin nuqtalarda ham egrilik noldan farqli bo'ladi

Shuning uchun, M nuqtaga yaqin nuqtalarda $\dot{\vec{r}}$ va $\ddot{\vec{r}}$ vektorlar o'zaro nokollinear bo'ladi. Demak, har bir nuqtadan yagona yopishma tekislik o'tadi. Agar $\vec{\beta}(s_0)$, $\vec{\beta}(s_0 + \Delta s)$ - vektorlar M va N nuqtadagi yopishma tekislikka perpendikulyar birlik vektorlar (ya'ni birlik binormal vektorlar) bo'lsa,

$$2 \sin \frac{\Delta\varphi}{2} = \left| \vec{\beta}(s_0 + \Delta s) - \vec{\beta}(s_0) \right|$$

tenglik o'rinli bo'ladi.

Shuning uchun

$$\frac{|\vec{\beta}(s_0 + \Delta s) - \vec{\beta}(s_0)|}{\Delta s} = \frac{2 \sin \frac{\Delta\varphi}{2}}{\Delta s} = \frac{\sin \frac{\Delta\varphi}{2}}{\frac{\Delta\varphi}{2}} \cdot \frac{\Delta\varphi}{\Delta s}$$

tenglik o'rinli. Bu tenglikda $\Delta s \rightarrow 0$ limitga o'tib, $|\sigma| = \left| \dot{\vec{\beta}} \right|$ tenglikni hosil qilamiz. Binormal $\vec{\beta}$ vektor birlik vektor bo'lganligi uchun $\dot{\vec{\beta}} \perp \vec{\beta}$ bo'ladi. Agar $\vec{\tau}(s) = \dot{\vec{r}}(s)$ bo'lsa, $\vec{\nu} = \frac{\dot{\vec{r}}}{k}$ - birlik boshnormal vektor, $\vec{\tau}$ - birlik urinma vektor bo'ladi. Shuning uchun $\vec{\beta} = [\vec{\tau}, \vec{\nu}]$

bo'ladi. Demak, $\ddot{\vec{\beta}} = [\ddot{\vec{r}}, \dot{\vec{v}}] + [\ddot{\vec{r}}, \dot{\vec{v}}] = [\ddot{\vec{r}}, \dot{\vec{v}}]$, chunki $[\dot{\vec{r}}, \dot{\vec{v}}] = \vec{0}$. Bu

tenglikdan, $\ddot{\vec{\beta}} \perp \dot{\vec{r}}$ ekanligi kelib chiqadi. Demak, $\ddot{\vec{\beta}} \parallel \dot{\vec{v}}$. Shuning

uchun, $|\sigma| = \left| \left(\ddot{\vec{\beta}}, \dot{\vec{v}} \right) \right|$ tenglikni yoza olamiz. Bu tenglikka

$\dot{\vec{v}} = \frac{\dot{\vec{r}}}{k}$, $\ddot{\vec{\beta}} = \frac{[\ddot{\vec{r}}, \dot{\vec{r}}]}{k}$ ifodalarni qo'yib, $|\sigma| = \frac{|\dot{\vec{r}} \ddot{\vec{r}} \ddot{\vec{r}}|}{k^2}$ formulani hosil

qilamiz. Endi buralishni aniqlaylik. $\ddot{\vec{\beta}}$ vektor $\dot{\vec{v}}$ vektorga parallel bo'lganligi uchun egri chiziq bo'ylab harakat qilsak (s o'sa boshlaganda) yopishma tekislik urinma atrofida aylana boshlaydi. Agar yopishma tekislik buralishi yo'nalishi $\ddot{\vec{\beta}}$ dan $\dot{\vec{v}}$ ga yo'налgan bo'lsa, (+) ishora bilan aks holda esa (-) ishora bilan olib, $\sigma = \pm |\sigma|$ formula bo'yicha buralishni kiritamiz. $|\sigma|$ ning ifodasini hisobga olib

$$\sigma = - \frac{\dot{\vec{r}} \ddot{\vec{r}} \ddot{\vec{r}}}{k^2}$$

formulani hosil qilamiz.

Endi ixtiyoriy t parametr uchun buralishni hisoblash formulasini keltirib chiqaramiz. Buning uchun yoy uzunligi $S = S(t)$ parametr t ning funktsiyasi ekanligidan foydalanamiz.

Egri chiziq tenglamasi $\vec{r} = \vec{r}(s)$ bo'lsa,

$$\begin{aligned} \dot{\vec{r}} &= \vec{r}' \cdot \frac{dt}{ds}, \quad \ddot{\vec{r}} = \vec{r}'' \left(\frac{dt}{ds} \right)^2 + \vec{r}' \cdot \frac{d^2t}{ds^2}, \\ \ddot{\vec{r}} &= \vec{r}''' \left(\frac{dt}{ds} \right)^3 + 2\vec{r}'' \frac{dt}{ds} \cdot \frac{d^2r}{ds^2} + \vec{r}'' \frac{d^2r}{ds^2} \frac{dt}{ds} + \vec{r}' \frac{dt}{ds} \cdot \frac{d^3r}{ds^3} \end{aligned}$$

ifodalarni buralish formulasiga qo'ysak

$$\sigma = - \frac{\vec{r}' \ddot{\vec{r}} \ddot{\vec{r}}}{[\vec{r}', \ddot{\vec{r}}]^2}$$

formulani hosil qilamiz.

Agar birorta chiziqning buralishi hamma nuqtalarda nolga teng bo'lsa, u albatta yassi chiziq bo'ladi, ya'ni birorta tekislikda yotadi (isbotlang).

Yuqorida ko'rsatib o'tganimizdek, agar regulyar γ chiziq

$$\vec{r} = \vec{r}(t), \quad a < t < b$$

tenglama bilan berilib, har bir t uchun $\vec{r}'(t)$ va $\vec{r}''(t)$ vektorlar kollinear vektorlar bo'lmasa, γ chiziqning har bir nuqtasiga ortonormal sistemani tashkil qiluvchi uchta vektorni mos qo'yish mumkin. Bu uchlik birlik urinma vektor, birlik bosh normal vektor va birlik binormal vektorlardan iborat. Bu uchlikni Frene uchligi deb ataymiz. Hozir biz fazodagi orientatsiyani saqllovchi harakat regulyar chiziqni regulyar chiziqqa o'tkazishini va bunda Frene uchligi ham yana Frene uchligiga o'tishini isbotlaymiz.

Fazoda regulyar γ egri chiziq

$$\vec{\rho} = \vec{\rho}(t), \quad a < t < b$$

tenglama bilan, uning $F: R^3 \rightarrow R^3$ harakatdagi obrazi $F(\gamma)$

$$\vec{r} = \vec{r}(t), \quad a < t < b$$

tenglama bilan berilgan bo'lsin. Agar

$$\vec{\rho}(t) = \{x(t), y(t), z(t)\}$$

bo'lib, F harakat

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

matritsa va

$$a = \{a_1, a_2, a_3\}$$

vektor yordamida berilgan bo'lsa, $F(x, y, z)$ nuqtaning koordinatalari

$$x_1 = a_{11}x + a_{12}y + a_{13}z + a_1$$

$$y_1 = a_{21}x + a_{22}y + a_{23}z + a_2$$

$$z_1 = a_{31}x + a_{32}y + a_{33}z + a_3$$

ko'rinishda bo'ladi. Shuning uchun $\vec{r}(t)$ vektorning koordinatalari

$$x_1(t), y_1(t), z_1(t)$$

funksiyalar bo'lsa,

$$(x_1(t), y_1(t), z_1(t)) = F(x(t), y(t), z(t))$$

tenglikdan

$$\vec{r}'(t) = A\vec{\rho}'(t)$$

formula kelib chiqadi. Bu tenglikda $\vec{r}'(t)$ va $\vec{\rho}'(t)$ vektorlar ustun ko'rinishda yozilgan. Bu yerda A ortogonal matritsa bo'lgani uchun

$$|\vec{r}'(t)| = |A\vec{\rho}'(t)| = |\vec{\rho}'(t)|,$$

$$(\vec{r}'(t), \vec{r}''(t)) = (A\vec{\rho}'(t), A\vec{\rho}''(t)) = (\vec{\rho}'(t), \vec{\rho}''(t)),$$

$$[\vec{r}'(t), \vec{r}''(t)] = [A\vec{\rho}'(t), A\vec{\rho}''(t)] = [\vec{\rho}'(t), \vec{\rho}''(t)]$$

tengliklar o'rinli. Bu tengliklardan oxirgisi o'rinli bo'lishi uchun $\det A > 0$ shartni ham ya'ni F harakat oriyentatsiyani saqlashini talab qildik. Bu tengliklardan

$$\tau_1 = \frac{\vec{r}'}{|\vec{r}'|} = \frac{A\vec{\rho}'}{|A\vec{\rho}'|} = \frac{A\vec{\rho}'}{|\vec{\rho}'|} = A(\vec{\tau})$$

$$\vec{v}_1 = A(\vec{v}), \quad \vec{\beta}_1 = [A(\vec{\tau}), A(\vec{v})] = A(\vec{\beta})$$

formulalar hosil qilamiz. Bu formulalar γ chiziqning Frene uchligi F akslantirishda $F(\gamma)$ chiziqning Frene uchligiga o'tishini isbotlaydi.

Bu formulalardan orientatsiyani saqlovchi harakatda chiziqlarning egriligi va buralishi ham o'zgarmay qolishi kelib

chiqadi. Haqiqatdan, egrilik va buralish formulalaridan foydalanib,

$$k_1 = \frac{\left[\begin{matrix} \vec{r}', \vec{r}'' \end{matrix} \right]}{\left(\vec{r}'^2 \right)^{\frac{3}{2}}} = k = \frac{\left[\begin{matrix} \vec{\rho}', \vec{\rho}'' \end{matrix} \right]}{\left(\vec{\rho}'^2 \right)^{\frac{3}{2}}}$$

$$\sigma_1 = -\frac{\vec{r}' \cdot \vec{r}'' \cdot \vec{r}'''}{k_1^2} = \sigma = -\frac{\vec{\rho}' \cdot \vec{\rho}'' \cdot \vec{\rho}'''}{k^2}$$

tengliklarni hosil qilamiz.

§ 8. Frene formulari

Egri chiziq γ tabiiy parametr yordamida

$$\vec{r} = \vec{r}(s)$$

tenglama bilan berilgan bo'lsin. Agar M nuqta γ ning parametrning s_0 qiymatiga mos keluvchi nuqta bo'lsa, bu nuqtadan chiquvchi o'zaro ortogonal uchta vektor mavjudligini ko'rdik.

Bular, $\vec{\tau}(s_0)$ – birlik urinma vektor, $\vec{\nu}(s_0)$ – birlik bosh normal vektor, $\vec{\beta}(s_0)$ – birlik binormal vektorlardir. Egri chiziq γ ning M nuqta atrofidagi qismini tekshirishda M nuqtani koordinata boshi sifatida, $\vec{\tau}, \vec{\nu}, \vec{\beta}$ –vektorlarni koordinata o'qlarining yo'naltiruvchi vektorlar sifatida olaylik. Buning uchun, oldin $\vec{\tau}, \vec{\nu}, \vec{\beta}$ vektorlarning hosilalarini $\vec{\tau}, \vec{\nu}, \vec{\beta}$ vektorlar orqali ifodalaylik. Birinchidan, $\dot{\vec{r}} = \vec{r}' = k\vec{\nu}$ munosabatini bilamiz. Oldingi paragrafda $\dot{\vec{\beta}} = \sigma\vec{\nu}$ ni ko'rsatgan edik. Bularni va $\vec{\nu} = [\vec{\beta}, \vec{\tau}]$ ni hisobga olib $\dot{\vec{\nu}} = \left[\vec{\beta}, \vec{\tau} \right] + \left[\vec{\beta}, \vec{\tau} \right]$ dan $\vec{\nu} = -k\vec{\tau} - \sigma\vec{\beta}$ formulani hosil qilamiz.

Demak,

$$\begin{cases} \dot{\vec{\tau}} = k\vec{v} \\ \dot{\vec{v}} = -k\dot{\vec{\tau}} - \sigma\dot{\vec{\beta}} \\ \dot{\vec{\beta}} = \sigma\vec{v} \end{cases}$$

formulalari hosil qilamiz.

Endi $\vec{r}(s_0 + \Delta s)$ vektor-funksiyani Teylor qatoriga yoyaylik

$$\vec{r}(s_0 + \Delta s) = \vec{r}(s_0) + \dot{\vec{r}} \Delta s + \frac{\ddot{\vec{r}}(s_0)}{2} \Delta s^2 + \frac{\dddot{\vec{r}}(s_0)}{6} \Delta s^3 + \dots$$

M nuqta koordinata boshi bo'lganligi uchun $\vec{r}(s_0) = \vec{0}$. Bu tenglikda

$$\dot{\vec{r}} = \vec{\tau}, \quad \ddot{\vec{r}} = k\vec{v}, \quad \dddot{\vec{r}} = k\vec{v} - k\sigma\vec{\beta} - k^2\vec{\tau}$$

munosabatlarni hisobga olib,

$$\vec{r}(s_0 + \Delta s) = \left(\Delta s - \frac{k^2 \Delta s^2}{6} + \dots \right) \vec{\tau} + \left(\frac{k \Delta s^2}{2} + \frac{\sigma \Delta s^2}{6} + \dots \right) \vec{v} + \left(-\frac{k \sigma \Delta s^2}{6} + \dots \right) \vec{\beta}$$

tenglikni hosil qilamiz.

Endi x, y, z o'qlari mos ravishda $\vec{\tau}, \vec{v}, \vec{\beta}$ vektorlar yo'nalishlariga ega ekanligidan foydalanib

$$x = \Delta s - k \frac{\Delta s^2}{6} + \dots$$

$$y = k \cdot \frac{\Delta s}{2} + k \frac{\Delta s^2}{6} + \dots$$

$$z = -k\sigma \frac{\Delta s^2}{6} + \dots$$

tenglamalarni hosil qilamiz. Bu tenglamalarda faqat egrilik va buralish qatnashmoqda. Demak, chiziqni aniqlash uchun uning hamma nuqtalarida egrilik va buralishni bilishimiz yetarli.

Endi shu masalani muhokama qilaylik. Bizga parametrlangan regulyar γ egri chiziq berilgan bo'lsa, uning ixtiyoriy nuqtasida uchta $s(t)$, $k(t)$, $\sigma(t)$ funktsiyalar aniqlangan. Bu funktsiyalar uzluksiz va $k(t) > 0$, $s(t) > 0$, munosabatlar

o'rinlidir. Agar parametr sifatida yoy uzunligini olsak, funksiyalar soni 2 ta bo'ladi.

Teorema-14. *Ikkita regulyar egri chiziqlarning yoylari γ_1 va γ_2 mos ravishda*

$$\vec{r} = \vec{r}_1(t), \quad \vec{r} = \vec{r}_2(t),$$

tenglamalar yordamida berilib,

$$\int_a^t |\vec{r}_1'(t)| dt = \int_a^t |\vec{r}_2'(t)| dt$$

tenglik ixtiyoriy $t \in [a, b]$ uchun o'rinli bo'lsin. Bundan tashqari har bir $t \in [a, b]$ uchun $k_1(t) = k_2(t)$, $\sigma_1(t) = \sigma_2(t)$ tengliklar o'rinli bo'lsa, yagona $F: R^3 \rightarrow R^3$ harakat mavjud bo'lib,

$$F(\gamma_2) = \gamma_1$$

munosabat o'rinli bo'ladi.

Isbot. Bu chiziqlarning uzunliklari teng bo'lgani uchun

$$s_0 = \int_a^b |\vec{r}_1'(t)| dt = \int_a^b |\vec{r}_2'(t)| dt$$

belgilash kiritib, chiziqlar tenglamalarini tabiiy parametr yordamida yozamiz. Shunda ularning tenglamalari

$$\vec{\rho} = \vec{\rho}_1(s)$$

$$\vec{\rho} = \vec{\rho}_2(s)$$

ko'rinishda bo'ladi. Endi har bir chiziqda tabiiy parametrning $s = 0$ qiymatiga mos keluvchi nuqtalarini mos ravishda M_1 va M_2 bilan belgilaymiz. Bu nuqtalardagi Frene uchliklari mos ravishda, $\vec{\tau}_1(0)$, $\vec{\nu}_1(0)$, $\vec{\beta}_1(0)$ va $\vec{\tau}_2(0)$, $\vec{\nu}_2(0)$, $\vec{\beta}_2(0)$ vektorlardan iborat bo'ladi. Bu uchliklar fazoda bir xil oriyentatsiyalarni aniqlagani uchun shunday $F: R^3 \rightarrow R^3$ harakat mavjudki, u M_2 nuqtani M_1 nuqtaga, $\vec{\tau}_2(0)$, $\vec{\nu}_2(0)$, $\vec{\beta}_2(0)$ vektorlarni mos ravishda $\vec{\tau}_1(0)$, $\vec{\nu}_1(0)$, $\vec{\beta}_1(0)$ vektorlarga o'tkazadi. Biz $F(\gamma_2) = \gamma_1$ tenglikni isbotlaymiz. Buning uchun $F(\gamma_2(s))$ nuqtaning radius-vektorini

$\vec{\rho}(s)$ bilan belgilab, $\vec{\rho} = \vec{\rho}(s)$, $s \in [0, s_0]$ tenglama bilan aniqlangan regulyar egri chiziqlarning Frene uchligini $\{\vec{\tau}(s), \vec{\nu}(s), \vec{\beta}(s)\}$ bilan belgilaymiz.

Shunda biz $\vec{\tau}(0) = \vec{\tau}_1(0)$, $\vec{\nu}(0) = \vec{\nu}_1(0)$, $\vec{\beta}(0) = \vec{\beta}_1(0)$ tengliklarga ega bo'lamiz.

Harakatda vektorlarning skalyar ko'paytmasi saqlangani uchun

$k(s) = k_2(s)$, $\sigma(s) = \sigma_2(s)$
tengliklar o'rinli bo'ladi. Demak,
 $k(s) = k_1(s)$, $\sigma(s) = \sigma_1(s)$ tengliklar ham o'rinlidir.

Endi $\vec{\rho}(s) = \vec{\rho}_1(s)$ tenglikni isbotlash uchun

$$h(s) = (\vec{\tau}_1(s), \vec{\tau}(s)) + (\vec{\nu}_1(s), \vec{\nu}(s)) + (\vec{\beta}_1(s), \vec{\beta}(s))$$

funksiyani qaraymiz. Bu funksiya uchun $h(0) = 3$ tenglik o'rinli. Bu funksiyaning differensiallaymiz

$$h'(s) = (\dot{\vec{\tau}}_1(s), \vec{\tau}(s)) + (\vec{\tau}_1(s), \dot{\vec{\tau}}(s)) + (\dot{\vec{\nu}}_1(s), \vec{\nu}(s)) + (\vec{\nu}_1(s), \dot{\vec{\nu}}(s)) + (\dot{\vec{\beta}}_1(s), \vec{\beta}(s)) + (\vec{\beta}_1(s), \dot{\vec{\beta}}(s))$$

va Frene formulalaridan foydalanib,

$$\begin{aligned} h'(s) &= k_1(\vec{\nu}_1(s), \vec{\tau}(s)) + k(\vec{\tau}_1(s), \vec{\nu}(s)) - k_1(\vec{\tau}_1(s), \vec{\nu}(s)) - \sigma_1(\vec{\beta}_1(s), \vec{\nu}(s)) - \\ &- k(\vec{\nu}_1(s), \vec{\tau}(s)) - \sigma(\vec{\nu}_1(s), \vec{\beta}(s)) + \sigma_1(\vec{\nu}_1(s), \vec{\beta}(s)) + \sigma(\vec{\beta}_1(s), \vec{\nu}(s)) = \\ &= (k_1 - k)(\vec{\nu}_1(s), \vec{\tau}(s)) + (k - k_1)(\vec{\tau}_1(s), \vec{\nu}(s)) + (\sigma_1 - \sigma)(\vec{\nu}_1(s), \vec{\beta}(s)) - \\ &- (\sigma_1 - \sigma)(\vec{\beta}_1(s), \vec{\nu}(s)) \end{aligned}$$

tenglikni hosil qilamiz. Bu yerda $k = k_1$, $\sigma = \sigma_1$ bo'lgani uchun $h'(s) = 0$. Demak, $h(s) = h(0) = 3$ va $\vec{\tau}(s) = \vec{\tau}_1(s)$ tenglik o'rinli bo'ladi. Bundan $\vec{\rho}(s) = \vec{\rho}_1(s) + \vec{c}$ tenglikni olamiz bu yerda $\vec{c}$ - o'zgarmas vektor bo'lgani uchun $\vec{\rho}(0) = \vec{\rho}_1(0)$

tenglikdan $\vec{c} = \vec{0}$ munosabat kelib chiqadi. Shunday qilib, biz $F(\gamma_2) = \gamma_1$ munosabatni isbotladik.

Teorema-15. *Ikkita uzluksiz $f(s)$ va $g(s)$ funktsiyalar $[0; s_0]$ oraliqda aniqlangan va $f(s) > 0$ bo'lsa, tabiiy parametr yordamida parametrlangan regulyar egri chiziq mavjud bo'lib, uning egriligi hamda buralishi mos ravishda $f(s), g(s)$ funktsiyalarga tengdir.*

Isbot. Bizga M_0 nuqta va ortonormal $\{\vec{a}, \vec{b}, \vec{c}\}$ sistema berilgan bo'lsin. Biz $\vec{\tau}, \vec{\nu}, \vec{\beta}$ vektor funktsiyalarga nisbatan

$$\begin{aligned}\dot{\vec{\tau}} &= f\vec{\nu} \\ \dot{\vec{\nu}} &= -f\vec{\tau} - g\vec{\beta} \\ \dot{\vec{\beta}} &= g\vec{\nu}\end{aligned}\quad (1)$$

differensial tenglamalar sistemasini

$$\vec{\tau}(0) = \vec{a}, \quad \vec{\nu}(0) = \vec{b}, \quad \vec{\beta}(0) = \vec{c}$$

boshlang'ich shartlar bilan qaraylik. Differensial tenglamalar sistemasining yechimi mavjudligi va yagonaligi haqidagi teorema asosan bu sistemaning $[0, s_0]$ oraliqda aniqlangan yagona $\{\vec{\tau}(s), \vec{\nu}(s), \vec{\beta}(s)\}$ yechimi mavjud. Boshlang'ich shartlarga asosan $s=0$ bo'lganda bu uchlik ortonormal sistemani tashkil qiladi. Biz ixtiyoriy $s \in [0, s_0]$ uchun bu uchlikning ortonormal ekanligini ko'rsatamiz. Buning uchun $X(s)$ bilan birinchi satri $\vec{\tau}(s)$ vektordan, ikkinchi satri $\vec{\nu}(s)$ vektordan va uchinchi satri $\vec{\beta}(s)$ vektordan iborat matritsani belgilasak, (1) sistemani

$$X'(s) = A(s)X(s) \quad (2)$$

ko'rinishda yoza olamiz. Bu yerda

$$A(s) = \begin{pmatrix} 0 & f(s) & 0 \\ -f(s) & 0 & -g(s) \\ 0 & g(s) & 0 \end{pmatrix}$$

Endi $\bar{\tau}(s), \bar{\nu}(s), \bar{\beta}(s)$ vektorlarning ortonormal sistema ekanligini ko'rsatish uchun $X(s)$ matritsaning ortogonal matritsa ekanligini ko'rsatish yetarlidir. Demak, ixtiyoriy $s \in [0, S_0]$ uchun

$$X^T(s)X(s) = E$$

tenglikni isbotlashimiz zarur va yetarli. Bu yerda $X^T(s)$ – transponirlangan matritsa, E – birlik matritsadir.

Biz (2) tenglikdan

$$\frac{d}{ds}(X^T(s)) = X^T(s)A^T(s)$$

tenglikni olamiz. Bu tenglikni hisobga olib,

$$X^T(s)X(s)$$

ko'paytmani differensiallaymiz. Shunda

$$\begin{aligned} \frac{d}{ds}(X^T(s)X(s)) &= \frac{d}{ds}X^T(s)X(s) + X^T(s)\frac{d}{ds}X(s) = \\ &= X^T(s)A^T(s)X(s) + X^T(s)A(s)X(s) = X^T(s)[A^T(s) + A(s)]X(s) \end{aligned}$$

tenglikni hosil qilamiz. Bu tenglikda $A^T(s) = -A(s)$ munosabatni hisobga olib,

$$\frac{d}{ds}(X^T(s)X(s)) = 0$$

tenglikni hosil qilamiz. Demak, $X^T(s)X(s)$ o'zgarmas matritsa va $X^T(0)X(0) = E$ bo'lganligi uchun

$$X^T(s)X(s) = E$$

tenglik hammasi uchun o'rinlidir.

Shunday qilib, ixtiyoriy $s \in [0, s_0]$ uchun $\vec{\tau}(s)$, $\vec{\nu}(s)$, $\vec{\beta}(s)$ vektorlar ortonormal sistemani tashkil qiladi. Endi

$$\vec{\rho}(s) = \vec{\rho}_0 + \int_0^s \vec{\tau}(s) ds$$

tenglama bilan γ chiziqni aniqlaymiz. Bu yerda $\vec{\rho}_0$ vektor M_0 nuqtaning radius-vektori. Bu chiziq uchun

$$\dot{\vec{\rho}}(s) = \vec{\tau}(s), \quad |\vec{\tau}(s)| = 1$$

$$\ddot{\vec{\rho}}(s) = \dot{\vec{\tau}}(s) = f(s)\vec{\nu}(s)$$

$$k = |\ddot{\vec{\rho}}| = f(s)$$

bo'lganligi uchun

$$[\dot{\vec{\rho}}(s), \ddot{\vec{\rho}}(s)] \neq 0$$

munosabat kelib chiqadi. Demak, bu chiziq uchun buralish aniqlangan va

$$\begin{aligned} \sigma &= -\frac{\dot{\vec{\rho}}\ddot{\vec{\rho}}\ddot{\vec{\rho}}}{k^2} = -\frac{([\vec{\tau}, f\vec{\nu}], f'(s)\vec{\nu}(s) + f\dot{\vec{\nu}})}{k^2} = -\frac{([\vec{\tau}, f\vec{\nu}], f\dot{\vec{\nu}})}{k^2} = \\ &= -\frac{f^2([\vec{\tau}, \vec{\nu}] - f\vec{\tau} - g\vec{\beta})}{k^2} = \frac{f^2 g(\vec{\beta}, \vec{\beta})}{k^2} = g \end{aligned}$$

tenglik o'rinlidir. Demak, γ chiziq teorema tasdig'ini qanoatlantiradi. Agar M_0 nuqta o'rniga boshqa M nuqta olsak, biz teorema shartini qanoatlantiruvchi va M nuqtadan chiquvchi γ_M chiziqni hosil qilamiz. Lekin, teorema-12 ga ko'ra, $F: R^3 \rightarrow R^3$ harakat mavjud bo'lib, $F(\gamma_M) = \gamma$ bo'ladi. $\square$